Digital Data Transmission 2026–27

Universitat Pompeu Fabra Barcelona

Modality A: Midterm exam (sample 3)

- Each statement below concerns one part of a statement of the Midterm collection, and it may be worded differently from the collection. Clearly **select** (a) if the statement is true and (b) if it is false. Marks are +1 for a correct answer, −1 for a wrong answer, and 0 for no answer.
- No derivations or justifications are required, and none is marked.
- The exam **grade** is $R - W$, the number of right answers minus the number of wrong answers, over the 12 statements.
- No electronic devices and no printed or written material are allowed.
- The exam duration is **30 minutes**.

Name:

NIA or U:

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1. If $\phi(t)$ has energy E, then $\alpha\phi(t)$ has energy $|\alpha|E$ for any complex α.

(a) True
(b) False

2. The representation $\bar{x}(t) = \sum_{i=1}^{N} x_i\phi_i(t)$ of a signal $x(t)$ on an orthonormal set is the orthogonal projection of $x(t)$ onto the signal space, the signal of the space closest to $x(t)$, and it equals $x(t)$ exactly when $x(t)$ belongs to the space.

(a) True
(b) False

3. A rate or a capacity in bits per channel use is converted to bits per second by multiplying by the passband bandwidth W.

(a) True
(b) False

4. Nyquist's criterion: with a baseband waveform of bandwidth W, which occupies $|f| \le W$, the minimum symbol period without inter-symbol interference is $T = \frac{1}{W}$.

(a) True
(b) False

5. If $x(t) = \sum_{i=1}^{N} x_i\phi_i(t)$ with $\{\phi_i\}_{i=1}^{N}$ orthonormal, its energy is the sum of the squared moduli of its coefficients, $E = \sum_{i=1}^{N} |x_i|^2$.

(a) True
(b) False

6. If $x(t) = \sum_{i=1}^{N} a_i\phi_i(t)$ with unit-energy waveforms and independent zero-mean random coefficients, the average energy is $\mathrm{E}(E) = \sum_{i=1}^{N} \mathrm{E}(|a_i|^2)$ only when the waveforms are orthogonal.

(a) True
(b) False

7. In MATLAB, `randn` returns a real Gaussian sample of zero mean and unit variance, so σ `randn`$+\mu$ is Gaussian with mean μ and variance σ^2.

(a) True
(b) False

8. The binary entropy function $h_2(p)$ attains its maximum, 1 bit, at $p = 0$ and at $p = 1$.

(a) True
(b) False

9. A coded-modulation system with a constellation of M points can transmit reliably above $\log_2 M$ bits per channel use when the channel has no noise.

(a) True
(b) False

10. The channel law of the complex AWGN channel, $W(y|x) = \frac{1}{\pi\sigma^2} e^{-|y-x|^2/\sigma^2}$, depends on y only through the distance $|y - x|$, so all outputs at the same distance from x are equally likely.

(a) True
(b) False

11. For two equiprobable real symbols $\pm A$ in real Gaussian noise of variance σ^2, the ML detector is a threshold at 0 and $P_{\mathrm{e}} = Q\big(\frac{A}{\sigma}\big)$, with $Q(u) = \int_u^{\infty} \frac{e^{-t^2/2}}{\sqrt{2\pi}}\,\mathrm{d}t$.

(a) True
(b) False

12. A repetition code of length r followed by an M-point constellation has rate $r \log_2 M$ bits per channel use.

(a) True
(b) False